

Zakat Calculator

Project Documentation

Flutter training assignment

1. Project Overview

The Zakat Calculator is a Flutter application developed as a training assignment. It allows a user to enter relevant monetary assets and liabilities, calculate a zakat estimate using the configured nisab standard, and save calculations locally for later review.

The application focuses on calculation correctness, local data persistence, basic authentication, separation of calculation logic from the user interface, and automated unit testing.

2. Main Features

- **Local registration and login:** Users can create accounts and authenticate on the device.
- **Asset and liability entry:** Inputs cover gold, silver, cash, bank balances, business inventory, receivables, and debts.
- **Zakat calculation:** The calculation applies the configured nisab standard and a 2.5% rate, with the result shown as a breakdown.
- **Madhhab settings:** A setting controls the treatment of personal jewellery.
- **Calculation history:** Saved calculations can be reopened and are scoped to the logged-in user.
- **Automated tests:** Unit tests cover the required calculation cases, including boundary and edge cases.

3. Technology and Storage

The application is built with Flutter and Dart for Android. The calculation logic is kept in plain Dart so it can be tested independently of the widget tree. The decimal package is used for monetary arithmetic to avoid relying on binary floating-point values for money.

Hive is used for structured local records such as user-related data and calculation history. SharedPreferences stores lightweight settings and session-related values.

FlutterSecureStorage stores credential material separately from ordinary application data. The crypto package is used for salted password hashing, and UUIDs provide identifiers where needed.

4. Calculation Rules

The application calculates zakat on the monetary asset categories included in the assignment. Gold and silver values account for the entered purity; business inventory is valued using the entered market value; good receivables are included, while doubtful receivables are excluded by default and can be included through the relevant setting. Entered debts are deducted, and the zakatable base is not allowed to fall below zero.

The user can select the gold or silver nisab standard. The configured assignment conventions are 87.48 grams for gold and 612.36 grams for silver, multiplied by the user-entered price per gram. The calculation assumes that the lunar-year holding condition (hawl) has been met; it does not track dates or balances over time.

The zakat rate is 2.5%. Intermediate monetary values are not rounded; only the final zakat amount is rounded to two decimal places using half-up rounding.

5. Authentication and Security Limitations

Authentication is performed locally, without a remote authentication server. Password salts and hashes are stored through FlutterSecureStorage, but salted SHA-256 is a fast hashing algorithm and is less resistant to offline password guessing than purpose-built password-hashing algorithms such as PBKDF2 or bcrypt. Local authentication cannot independently protect the application against every form of device compromise, rooted-device access, or malicious access to application data. Because there is no backend, the application cannot provide server-side identity verification, account recovery, or cross-device authentication.

6. Testing

Automated unit tests are located in `test/zakat_calculator_test.dart`. The required test suite covers nisab boundary conditions, gold purity, debt deductions, receivables, madhhab-dependent jewellery treatment, business inventory, and empty or zero inputs.

Run the tests from the project root:

```
flutter test
```

7. Running the Application

Requirements: Flutter SDK, Dart SDK (included with Flutter), and an Android device or emulator.

From the project root, run:

```
flutter pub get
flutter run
```

8. Scope and Limitations

- Metal prices are entered manually; the app does not fetch live market prices.
- The app covers the monetary zakat scope specified in the assignment and excludes livestock, agricultural produce (ushr), buried treasure (rikaz), and Zakat al-Fitr.
- Hawl tracking, multiple currencies, cloud synchronization, remote authentication, password reset, and email verification are not implemented.
- The calculator follows the assignment's specified simplifications. Questions involving disputed jurisprudential cases should be referred to a qualified scholar.

9. Project Status

The project documentation describes the intended implemented scope of the training assignment. Before submission, confirm that the latest source code, test results, and repository history match the statements in this report.